Bosons

$$\psi_{\{l_i\}}(x_1, \dots x_N) = \sqrt{\frac{N_1! \dots N_n!}{N!}} \sum_P{}' \psi_{l_1}(x_1) \dots \psi_{l_N}(x_N)$$

sum over permutations (P) of indices $\{l_i\}$

$$\sum_{i=1}^{n} N_i = N$$

Particles are identical:
and hence it does not matter which of them occupies which state.
All that matters is to know which state is occupied.
$\Rightarrow$ formulate everything is <u>space of occupation numbers</u> (Fock space)

$$\Phi(N_1, \dots, N_i, \dots N_n) = \Phi(\{N_i\})$$

OR, $|\Phi\rangle = |\{N_i\}\rangle$

- States with different sets $\{N_i\}$ are orthogonal

$$\langle \{N_i'\} | \{N_i\} \rangle = 0.$$

- State $|\Phi\rangle$ itself is normalized

$$\langle \{N_i\} | \{N_i\} \rangle = 1.$$

To move between states with different $\{N\}$ sets we introduce <u>creation</u> a_i^+ and <u>annihilation</u> a_i operators

<u>annihilation</u> $\hat{a}_i \, \Phi(N_1, \dots, N_i, \dots) = \sqrt{N_i} \, \Phi(N_1, \dots, N_i - 1, \dots)$

removes particle from state φ_i

$$\langle N_i - 1 | \hat{a}_i | N_i \rangle = \sqrt{N_i}$$

Hermitian conjugate $\left(\langle N_i - 1 | \hat{a}_i | N_i \rangle \right)^+ = \langle N_i | \hat{a}_i^+ | N_i - 1 \rangle$

<u>creation</u> $\hat{a}_i^+ \phi(N_1, \ldots, N_i \ldots) = \sqrt{N_i+1}\, \phi(N_1, \ldots, N_i+1 \ldots)$

<u>adds</u> particle to state φ_i

$$\hat{a}_i^+ \hat{a}_i \phi(N_1 \ldots N_i \ldots) = \hat{a}_i^+ \sqrt{N_i}\, \phi(N_1 \ldots N_i - 1 \ldots)$$

$$= \sqrt{N_i} \sqrt{N_i - 1 + 1}\, \phi(N_1 \ldots N_i \ldots) = N_i\, \phi(N_1 \ldots N_i \ldots)$$

$\Rightarrow$ $\hat{a}_i^+ \hat{a}_i$ has eigenvalue N_i (# particles in state φ_i)

$\Rightarrow$ $\hat{N}_i = \hat{a}_i^+ a_i$ = particle number operator.

Its hermitian $\hat{N}^+ = (a^+ a)^+ = a^+ (a^+)^+ = a^+ a = \hat{N}$.

Then $\hat{a}_i \hat{a}_i^+ \phi(N_1 \ldots N_i \ldots) = \hat{a}_i \sqrt{N_i+1}\, \phi(N_1 \ldots N_i+1 \ldots)$

$$= \sqrt{N_i+1} \sqrt{N_i+1}\, \phi(N_1 \ldots N_i \ldots) = (N_i+1)\, \phi(N_1 \ldots N_i \ldots)$$

$\Rightarrow$ $\hat{a}_i \hat{a}_i^+$ has eigenvalue $N_i + 1$

$$\Rightarrow \hat{a}_i \hat{a}_i^+ - \hat{a}_i^+ \hat{a}_i = N_i + 1 - N_i = 1.$$

$$[\hat{a}_i, \hat{a}_i^+] = 1.$$

Consider now $\hat{a}_i^+ \hat{a}_j \phi(\ldots N_i \ldots N_j \ldots) = \sqrt{N_i+1} \sqrt{N_j}\, \phi(N_i+1, N_j-1)$
$(i \neq j)$

$$\hat{a}_j \hat{a}_i^+ \phi(N_i \ldots N_j) = \sqrt{N_j} \sqrt{N_i+1}\, \phi(N_i+1, N_j-1)$$

$$\Rightarrow [\hat{a}_j, \hat{a}_i^+] = 0 \quad \text{e.g. } \hat{a}_j \hat{a}_i^+ = \hat{a}_i^+ \hat{a}_j$$

Bosonic w.f. is symmetric with respect to particle permutations.

Thus $[\hat{a}_j, \hat{a}_i^+] = \delta_{ij}$ in general. $\delta_{ij} = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$

And $[\hat{a}_j, \hat{a}_i] = [\hat{a}_j^+, \hat{a}_i^+] = 0$ always

Hence normalized wave function is

$$\phi(N_1, \ldots, N_i, \ldots) = \frac{(\hat{a}_1^+)^{N_1}}{\sqrt{N_1!}} \cdots \frac{(\hat{a}_i^+)^{N_i}}{\sqrt{N_i!}} \cdots \phi_0(0, 0, 0 \ldots)$$

where $\phi_0(0, 0, 0 \ldots)$ = is the <u>vacuum</u> w.f. with <u>no</u> particles $\rightarrow$ all occupation numbers $\{N_i\} = 0$

By def, $\hat{a}_i \phi_0(0, 0, 0 \ldots 0) = 0$.

<u>Fermions</u> $N_i = 0$ or $1 \Rightarrow N_i! = 1$.

But now we have to deal with fully antisymmetric w.f.

$$\Psi_{\{l_i\}}(x_1 \dots x_n) = \frac{1}{\sqrt{N!}} \begin{vmatrix} \psi_{l_1}(x_1) & \dots & \psi_{l_N}(x_1) \\ \vdots & & \vdots \\ \psi_{l_1}(x_n) & & \psi_{l_N}(x_N) \end{vmatrix}$$

such that $\psi(x_1 \dots x_i \dots x_j \dots) = -\psi(x_1 \dots x_j \dots x_i \dots)$

$\Rightarrow$ must have the same property in occupation number space (Fock space)

$$(*) \begin{cases} \phi(N_1 \dots N_i \dots N_j \dots) = -\phi(N_1 \dots N_j \dots N_i \dots) \\ \text{when } N_i = N_j = 1 \text{ (both states are occupied)} \end{cases}$$

$\Downarrow$

meaning that both states φ_i, φ_j appear in the Slater w.f.

We now define (same as for bosons, with $N_i = 0, 1$ only)

$$\phi(N_1 \dots N_i \dots) = (\hat{a}_1^+)^{N_1} \dots (\hat{a}_i^+)^{N_i} \dots \phi(0, 0, 0 \dots 0)$$

and deduce properties $\hat{a}_i$, $\hat{a}_i^+$ must have in order to satisfy $(*)$.

For state with $N_i = N_j = 1$ and all other $N_k = 0$, eq $(*)$ says:

$$0 = \phi(0 \dots 1_i \dots 1_j \dots 0) + \phi(0 \dots 1_j \dots 1_i \dots 0)$$

$$= \hat{a}_i^+ \hat{a}_j^+ \phi_0(0) + \hat{a}_j^+ \hat{a}_i^+ \phi_0(0)$$

Thus $\hat{a}_i^+ \hat{a}_j^+ + \hat{a}_j^+ \hat{a}_i^+ = 0 = \{\hat{a}_i^+, \hat{a}_j^+\}$

Note that here we can have $i=j$ too because

$\hat{a}_i^+ \hat{a}_i^+ \phi_0(0) = 0$ (can not have 2 fermi particles in the same state φ_i)

Next:

$$\hat{a}_i^+ \phi(N_1 \dots N_{i-1}, 0_i, N_{i+1} \dots) =$$

$$= \hat{a}_i^+ (\hat{a}_1^+)^{N_1} \dots (\hat{a}_{i-1}^+)^{N_{i-1}} (\hat{a}_{i+1}^+)^{N_{i+1}} \dots \phi_0(0 \dots 0)$$

But $\hat{a}_i^+ (\hat{a}_1^+)^{N_1} = \begin{cases} (N_1=1) \ \hat{a}_i^+ \hat{a}_1^+ = -\hat{a}_1^+ \hat{a}_i^+ \\ (N_1=0) \ \hat{a}_i^+ (\hat{a}_1^+)^0 = \hat{a}_i^+ = (\hat{a}_1^+)^0 \hat{a}_i^+ \end{cases}$

$$\Rightarrow (\hat{a}_i^+)(\hat{a}_1^+)^{N_1} = (-1)^{N_1} (\hat{a}_1^+)^{N_1} \hat{a}_i^+$$

$$(\hat{a}_1^+)^{N_1} (-1)^{N_1} \hat{a}_i^+ (\hat{a}_2^+)^{N_2} \dots (\hat{a}_{i-1}^+)^{N_{i-1}} (\hat{a}_{i+1}^+)^{N_{i+1}} \phi_0$$

$$= (-1)^{N_1 + N_2 + \dots + N_{i-1}} (\hat{a}_1^+)^{N_1} (\hat{a}_2^+)^{N_2} \dots (\hat{a}_{i-1}^+)^{N_{i-1}} \hat{a}_i^+ \cdot$$

$$\cdot (\hat{a}_{i+1}^+)^{N_{i+1}} \dots \phi_0$$

$$\Rightarrow \hat{a}_i^+ \phi(N_1, \dots, N_{i-1}, 0_i, N_{i+1} \dots) =$$

$$= (-1)^{\sum_{l=1}^{i-1} N_l} \phi(N_1, \dots, N_{i-1}, 1_i, N_{i+1} \dots)$$

$$\Rightarrow \langle 1_i | \hat{a}_i^+ | 0_i \rangle = (-1)^{\sum_{l=1}^{i-1} N_l}$$

Conjugate $\left(\langle 1_i | \hat{a}_i^+ | 0_i \rangle\right)^+ = \langle 0_i | \hat{a}_i | 1_i \rangle = (-1)^{\sum_1^{i-1} N_\ell}$

$\Rightarrow$ destruction operator is defined as

$$\hat{a}_i \, \phi(N_1 \ldots, 1_i, N_{i+1} \ldots) = (-1)^{\sum_{\ell=1}^{i-1} N_\ell} \phi(N_1 \ldots, 0_i, N_{i+1})$$

Now consider

$$\hat{a}_i^+ \hat{a}_i | 0_i \rangle = 0 \, | 0_i \rangle$$

$$\hat{a}_i^+ \hat{a}_i | 1_i \rangle = 1 \, | 1_i \rangle$$

$$\hat{a}_i \hat{a}_i^+ | 0_i \rangle = 1 \, | 0_i \rangle$$

$$\hat{a}_i \hat{a}_i^+ | 1 \rangle = 0 \, | 1_i \rangle$$

OR: $\left(\hat{a}_i^+ \hat{a}_i + \hat{a}_i \hat{a}_i^+\right) | 0_i \rangle = 1 \, | 0_i \rangle$

$$\left(\hat{a}_i^+ \hat{a}_i + \hat{a}_i \hat{a}_i^+\right) | 1_i \rangle = 1 \, | 1_i \rangle$$

$\Rightarrow$ $\hat{a}_i \hat{a}_i^+ + \hat{a}_i^+ \hat{a}_i = 1$ as an operator identity.

Together with $\hat{a}_j \hat{a}_i^+ + \hat{a}_i^+ \hat{a}_j = 0$ for $i \neq j$

from the next page, we have finally

$$\hat{a}_j \hat{a}_i^+ + \hat{a}_i^+ \hat{a}_j \equiv \{\hat{a}_j, \hat{a}_i^+\} = \delta_{ij}$$

Fermion operators $\hat{a}_j$, $\hat{a}_i^+$ anticommute!

let $j<i$

$A_1 \rightarrow \hat{a}_j \hat{a}_i^+ \phi(\ldots N_j \ldots N_i \ldots) =$ (must be $N_j = 1, N_i = 0$)

$$= \hat{a}_j (-1)^{\sum_{l=1}^{i-1} N_l} \phi(\ldots 1_j \ldots 1_i \ldots)$$

$$= (-1)^{\sum_{k=1}^{j-1} N_k} (-1)^{\sum_{l=1}^{i-1} N_l} \phi(\ldots 0_j \ldots 1_i \ldots)$$

this sum has $\underline{N_j = 1}$ in it.

$A_2 \rightarrow \hat{a}_i^+ \hat{a}_j \phi(\ldots 1_j \ldots 0_i \ldots) =$

$$= \hat{a}_i^+ (-1)^{\sum_{k'=1}^{j-1} N_{k'}} \phi(\ldots 0_j \ldots 0_i) =$$

$$= (-1)^{\sum_{k'}^{j-1} N_{k'}} (-1)^{\sum_{l'=1}^{i-1} N_{l'}} \phi(\ldots 0_j \ldots 1_i \ldots)$$

this sum has $\underline{N_j = 0}$ in it.

$\Rightarrow$ the two operations (A_1 and A_2) have different signs!

$$(-1)^{\sum_{l=1}^{i-1} N_l} = (-1)^{N_1 + N_2 + \ldots + \overset{=1}{N_j} + \ldots}$$

$$(-1)^{\sum_{l'=1}^{i-1} N_{l'}} = (-1)^{N_1 + N_2 + \ldots + \overset{=0}{N_j} +}$$

$$\Rightarrow \hat{a}_j \hat{a}_i^+ \phi(N_j \ldots N_i) = - \hat{a}_i^+ \hat{a}_j \phi(\ldots N_j \ldots N_i)$$

$$\Rightarrow \hat{a}_j \hat{a}_i^+ + \hat{a}_i^+ \hat{a}_j = 0 \text{ for } i \neq j.$$

Number operator $\hat{N}_i = \hat{a}_i^+ \hat{a}_i$ counts particles in <u>i-th</u> state

Total number $\hat{N} = \sum_i \hat{N}_i = \sum_i \hat{a}_i^+ \hat{a}_i$

Similarly, ~~any~~ for any operator $\hat{A}$, which ~~[illegible]~~ has eigenvalues A_i in <u>i-th</u> state, we can write operator identity:

$$\hat{A} = \sum_i A_i \hat{a}_i^+ \hat{a}_i$$

<u>Example</u>: free particles → eigenstate of free particle Hamiltonian are plane waves with momentum $\vec{p} = \hbar \vec{k}$, $e^{i\vec{k}\cdot\vec{r}}$, ⇒ states are numerated by $\vec{p}$:

$$\Rightarrow \hat{N} = \sum_p \hat{a}_p^+ \hat{a}_p = \sum_p \hat{N}_p$$

Total momentum $\vec{P} = \sum_p \vec{p}\, \hat{a}_p^+ \hat{a}_p$

Average value $\langle \vec{P} \rangle = \langle \Phi(N_p) | \vec{P} | \Phi(N_p) \rangle =$

$$= \sum_p \vec{p}\, N_{\vec{p}}$$

$$N_{\vec{p}} = \langle \hat{N}_{\vec{p}} \rangle$$

The same for kinetic energy operator:

$$\hat{T} = \sum_p \frac{\vec{p}^2}{2m} \hat{a}_p^+ \hat{a}_p = \sum_k \frac{\hbar^2 \vec{k}^2}{2m} \hat{a}_k^+ \hat{a}_k$$

How to handle potential energy?

$$\hat{U} = \sum_a U(\vec{r}_a)$$ where $\vec{r}_a$ = coordinate of <u>a</u>-th particle

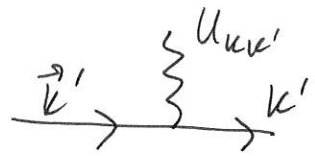
Need to find matrix elements of $\hat{U}(\vec{r}_a)$ between different momentum eigenstates $\varphi_\alpha(\vec{r}) = \frac{1}{\sqrt{V}} e^{i\vec{k}_\alpha \vec{r}}$ where different $\vec{k}_\alpha$ numerate different plane waves.

$$U_{\alpha\beta} = \langle \varphi_\alpha | \hat{U} | \varphi_\beta \rangle = \frac{1}{V} \int d^3\vec{r}\; e^{-i\vec{k}_\alpha \cdot \vec{r}}\, U(r)\, e^{i\vec{k}_\beta \vec{r}}$$

$$= \frac{1}{V} \int d^3\vec{r}\; e^{i(\vec{k}_\beta - \vec{k}_\alpha)\cdot \vec{r}}\, U(\vec{r})$$

More convenient to denote it as $U_{kk'} = \frac{1}{V} \int d\vec{r}\, e^{i(k'-k)r}\, U(r)$

Then $\hat{U} = \sum_{k,k'} U_{kk'}\, \hat{a}^+_k\, \hat{a}_{k'}$



$\left\{ \begin{array}{l} \text{particle with } \vec{k}' \text{ is destroyed} \\ \text{by external potential } U_{kk'}, \\ \text{particle with } \vec{k} \text{ is created} \end{array} \right.$

If $U_{kk'} = U_k\, \delta_{kk'}$ is diagonal in $\vec{k}$-basis, then $\hat{U} = \sum_k U_k\, \hat{a}^+_k\, \hat{a}_k$

K.E. is such an example because

$$T_{kk'} = \frac{1}{V} \int d^3\vec{r}\; e^{-i\vec{k}\cdot\vec{r}} \left(-\frac{\hbar^2}{2m} \vec{\nabla}_r^2 \right) e^{i\vec{k}'\cdot\vec{r}} =$$

$$= \frac{1}{V} \int d^3\vec{r}\; e^{-i\vec{k}\cdot\vec{r}}\, \frac{\hbar^2 (k')^2}{2m} e^{i\vec{k}'\cdot\vec{r}}$$

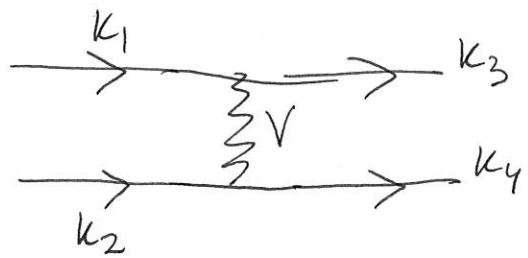
$$= \frac{\hbar^2 (k')^2}{2m} \frac{1}{V} \int d^3\vec{r}\; e^{i\vec{r}(\vec{k}'-\vec{k})} = \frac{\hbar^2 (k')^2}{2m} \delta_{k,k'}$$

Potential energy of pair-wise interactions between particles:

$$\hat{V} = \frac{1}{2} \sum_{a,b}{}' \hat{v}(\vec{r}_a - \vec{r}_b)$$

(sum over ~~all~~ particles with coordinates $\vec{r}_a$ and $\vec{r}_b$) ($a \neq b$, factor $\frac{1}{2}$ compensates over-counting)

This is <u>two-particle</u> operator



$$\Rightarrow \hat{V} = \frac{1}{2} \sum_{k_1+k_2=k_3+k_4} V_{k_4 k_3 k_2 k_1} \hat{a}^+_{k_4} \hat{a}^+_{k_3} \cdot \hat{a}_{k_2} \hat{a}_{k_1}$$

$$V_{k_4 k_3 k_2 k_1} = \int d^3\vec{r}_a \, d^3\vec{r}_b \left(\frac{1}{\sqrt{V}} e^{i\vec{k}_3\vec{r}_a}\right)^* \left(\frac{1}{\sqrt{V}} e^{i\vec{k}_4\vec{r}_b}\right)^*$$

$$\cdot \hat{v}(\vec{r}_a - \vec{r}_b) \cdot \frac{1}{\sqrt{V}} e^{i\vec{k}_2\vec{r}_b} \frac{1}{\sqrt{V}} e^{i\vec{k}_1\vec{r}_a} =$$

$$\begin{cases} \vec{R} = \frac{1}{2}(\vec{r}_a + \vec{r}_b) \\ \vec{r} = \vec{r}_a - \vec{r}_b \end{cases} \quad \begin{cases} \vec{r}_a = \vec{R} + \frac{\vec{r}}{2} \\ \vec{r}_b = \vec{R} - \frac{\vec{r}}{2} \end{cases}$$

$$= \frac{1}{V^2} \int d\vec{R}\, d\vec{r} \; e^{i(\vec{R}+\frac{\vec{r}}{2})(\vec{k}_1-\vec{k}_3)} \, e^{i(\vec{R}-\frac{\vec{r}}{2})(\vec{k}_2-\vec{k}_4)} \, v(\vec{r})$$

$$= \underbrace{\frac{1}{V} \int d\vec{R}\, e^{i\vec{R}(\vec{k}_1+\vec{k}_2-\vec{k}_3-\vec{k}_4)}}_{\delta_{\vec{k}_1+\vec{k}_2,\, \vec{k}_3+\vec{k}_4}} \; \frac{1}{V} \int d\vec{r}\, e^{i\frac{\vec{r}}{2}(\vec{k}_1-\vec{k}_2-\vec{k}_3+\vec{k}_4)} \cdot v(\vec{r})$$

conservation of momentum

Slightly different view of pages 38-40:

Start $\hat{N} = \sum_k \hat{a}_k^+ \hat{a}_k$.

Define $\hat{\psi}$-operators such that

$$\begin{cases} \hat{\psi}(\vec{r}) = \sum_k \hat{a}_k \, \varphi_k(\vec{r}) \\ \hat{\psi}^+(\vec{r}) = \sum_k \hat{a}_k^+ \, \varphi_k^*(\vec{r}) \end{cases}$$

where $\varphi_k(\vec{r})$ is full orthonormal set of single-particle w.f. (used to introduce $\hat{a}_k$)

$$\varphi_k(\vec{r}) = \frac{1}{\sqrt{V}} e^{i\vec{k}\cdot\vec{r}}$$

Consider now

$$\int d\vec{r} \, \hat{\psi}^+(r) \hat{\psi}(r) = \int d\vec{r} \sum_{kk'} \hat{a}_{k'}^+ \hat{a}_k \, \varphi_{k'}^*(\vec{r}) \varphi_k(\vec{r})$$

But by definition $\int d\vec{r} \, \varphi_{k'}^*(\vec{r}) \varphi_k(\vec{r}) = \delta_{k',k}$

$$\Rightarrow \int d\vec{r} \, \hat{\psi}^+(r) \hat{\psi}(r) = \sum_k \hat{a}_k^+ \hat{a}_k = \hat{N}$$

$\Rightarrow$ <u>density</u> operator is $\hat{\rho}(\vec{r}) = \hat{\psi}^+(\vec{r}) \hat{\psi}(\vec{r})$

Then single-particle P.E.

$$\hat{U} = \sum_a \hat{u}(\vec{r}_a) = \int \hat{\rho}(\vec{r})\, \hat{u}(\vec{r})\, d\vec{r}$$

(simply: pot. energy at point $\vec{r}$ <u>times</u> density at $\vec{r}$)

$$\hat{U} = \int d\vec{r}\, u(\vec{r})\, \hat{\psi}^+(r)\, \hat{\psi}(r) = \int d\vec{r} \sum_{kk'} \hat{a}^+_k \hat{a}_{k'}\, \varphi^*_k(\vec{r})\varphi_{k'}(\vec{r}) \cdot u(\vec{r})$$

$$= \sum_{kk'} \hat{a}^+_k \hat{a}_{k'} \int d\vec{r}\, \frac{1}{V} e^{-i\vec{k}\cdot\vec{r}} e^{i\vec{k}'\cdot\vec{r}}\, u(\vec{r}) =$$

$$= \sum_{kk'} u_{kk'}\, \hat{a}^+_k \hat{a}_{k'} = \sum_{kq} u_q\, \hat{a}^+_{k-q} \hat{a}_k$$

where again $u_{kk'} = \frac{1}{V} \int d\vec{r}\, e^{i(\vec{k}'-\vec{k})\vec{r}}\, u(\vec{r})$

~~depends~~ depends only on $\vec{k}' - \vec{k} = -\vec{q}$

$$u_{kk'} = u_q = \frac{1}{V} \int d\vec{r}\, e^{-i\vec{q}\cdot\vec{r}}\, u(\vec{r})$$

is just Fourier component of $u(\vec{r})$.

Commutation relations of ψ:

$$\{\hat{\psi}(\vec{r}), \hat{\psi}(\vec{r}')\} = \sum_{kk'} \varphi_k(\vec{r})\varphi_{k'}(\vec{r}')\, \{\hat{a}_k, \hat{a}_{k'}\} = 0$$

while $\{\hat{\psi}(\vec{r}), \hat{\psi}^+(\vec{r}')\} = \sum_{kk'} \varphi_k(\vec{r})\varphi^*_{k'}(\vec{r}')\, \underbrace{\{\hat{a}_k, \hat{a}^+_{k'}\}}_{=\delta_{k,k'}} =$

$= \sum_k \varphi_k(\vec{r})\varphi_k(\vec{r}') = \delta(\vec{r}-\vec{r}')$ due to completeness of the basis set $\{\varphi_k(\vec{r})\}$!

$\int \frac{d\vec{k}}{(2\pi)^3} e^{i\vec{k}(\vec{r}-\vec{r}')}$

Interaction energy

$$\hat{V} = \frac{1}{2}\sum_{a,b} v(\vec{r}_a - \vec{r}_b) = \frac{1}{2}\int d\vec{r}_1\, d\vec{r}_2\, v(\vec{r}_1 - \vec{r}_2)\, \hat{\rho}(\vec{r}_1)\hat{\rho}(\vec{r}_2)$$

$$= \frac{1}{2}\int d\vec{r}_1\, d\vec{r}_2\, v(\bar{r}_1 - \bar{r}_2)\, \frac{1}{V^2}\sum_{k_1 k_2 k_3 k_4} \hat{a}^+_{k_1}\hat{a}_{k_2}\hat{a}^+_{k_3}\hat{a}_{k_4} \cdot$$

$$\cdot\, e^{-i\bar{k}_1\bar{r}_1 + i\bar{k}_2\bar{r}_1 - i\bar{k}_3\bar{r}_2 + i\bar{k}_4\bar{r}_2}$$

$$\begin{cases} \vec{R} = \frac{1}{2}(\vec{r}_1 + \vec{r}_2) \\ \vec{r} = \vec{r}_1 - \vec{r}_2 \end{cases} \Rightarrow \begin{cases} \vec{r}_1 = \vec{R} + \vec{r}/2 \\ \vec{r}_2 = \vec{R} - \vec{r}/2 \end{cases}$$

let $V_{1234} = \frac{1}{2V^2}\underbrace{\int d\vec{R}\, e^{i\vec{R}\cdot(\bar{k}_1 - \bar{k}_2 + \bar{k}_3 - \bar{k}_4)}}_{V\delta(\bar{k}_1 - \bar{k}_2 + \bar{k}_3 - \bar{k}_4)} \int d\vec{r}\, v(\vec{r})\, e^{i\frac{\vec{r}}{2}(-\bar{k}_1 + \bar{k}_2 - \bar{k}_3 + \bar{k}_4)}$

Momentum conservation

$$V\delta(\bar{k}_1 - \bar{k}_2 + \bar{k}_3 - \bar{k}_4) \Rightarrow \bar{k}_1 - \bar{k}_2 = \bar{k}_4 - \bar{k}_3$$

$$= \frac{1}{2V}\int d\vec{r}\, v(\vec{r})\, e^{i\vec{r}\cdot(\vec{k}_4 - \vec{k}_3)} \equiv \frac{1}{2V}\, v(\vec{k}_4 - \vec{k}_3)$$

However it is not correct to write

$$\hat{V} = \frac{1}{2V}\sum_{\substack{k_1 k_2 k_3 k_4 \\ (k_1 + k_3 = \cancel{k_2} + k_4)}} v(\vec{k}_4 - \vec{k}_3)\, \hat{a}^+_{k_1}\hat{a}_{k_2}\hat{a}^+_{k_3}\hat{a}_{k_4}$$

because then V will result in a finite contribution from the state $|\vec{p}\rangle = \hat{a}^+_p|0\rangle$ with just <u>one</u> particle with momentum $\vec{p}$

Indeed: $\hat{V}|p\rangle = \frac{1}{2V}\sum_{(k_i)} v(\vec{k}_4-\vec{k}_3)\, \hat{a}^+_{k_1}\hat{a}_{k_2}\hat{a}^+_{k_3}\hat{a}_{k_4}\left(\hat{a}^+_p|0\rangle\right)$

But $\hat{a}^+_{k_1}\hat{a}_{k_2}\hat{a}^+_{k_3}\hat{a}_{k_4}\hat{a}^+_p|0\rangle =$

$$= \hat{a}^+_{k_1}\hat{a}_{k_2}\hat{a}^+_{k_3}|0\rangle\,\delta_{k_4,p} = \hat{a}^+_{k_1}|0\rangle\,\delta_{k_2,k_3}\,\delta_{k_4,p}$$

$$= \delta_{k_2,k_3}\,\delta_{k_4,p}\,|\vec{k}_1\rangle$$

$$\Rightarrow \hat{V}|\vec{p}\rangle = \frac{1}{2V}\sum_{\substack{k_1k_2k_3k_4\\(k_1+k_3=k_2+k_4)}} \delta_{k_2,k_3}\,\delta_{k_4,p}\,\vec{v}(k_4-k_3)\,|\vec{k}_1\rangle$$

$$= \frac{1}{2V}\sum_{\substack{k_1\,k_2\\(k_1=k_4=p)}} \vec{v}(\vec{p}-\vec{k}_2)\,|\vec{k}_1\rangle = \frac{1}{2V}\sum_{k_2}\vec{v}(\vec{p}-\vec{k}_2)\cdot|\vec{p}\rangle$$

But Two-particle interaction cannot affect one single particle state!

normal ordering [Simple fix: re-arrange operators so that all <u>annihilation</u> ones are to the right of <u>creation</u>

$$\hat{V} = \frac{1}{2V}\sum_{k_1k_2k_3k_4} v(\vec{k}_4-\vec{k}_3)\,\hat{a}^+_{k_1}\hat{a}^+_{k_3}\hat{a}_{k_4}\hat{a}_{k_2}$$

Check overall sign by considering the simplest case of constant $v(\vec{r}_a-\vec{r}_b) = v_0$ ($\vec{r}$-independent!)

Then $V = \frac{1}{2}\sum_{\substack{a,b\\(a\neq b)}} v_0 = \frac{1}{2}N(N-1)$ where N = total number of particles

But $v(\bar{k}_4-\bar{k}_3) = \int d\vec{r}\, v_0\, e^{i\vec{r}(\vec{k}_4-\vec{k}_3)} = \overset{\text{volume}}{V}\,\delta(\bar{k}_4-\bar{k}_3)\,v_0$

$$\Rightarrow \hat{V} = \frac{1}{2V}\sum_{(k_1+k_3=k_2+k_4)} V v_0\, \delta_{k_4,k_3}\, \hat{a}^+_{k_1}\hat{a}^+_{k_3}\hat{a}_{k_3}\hat{a}_{k_1}$$

$\Rightarrow k_1 = k_2$

$$= \frac{1}{2} v_0 \sum_{k_1 k_3} \hat{a}^+_{k_1}\hat{a}^+_{k_3}\underbrace{\hat{a}_{k_3}\hat{a}_{k_1}}_{-\hat{a}_{k_1}\hat{a}_{k_3}} =$$

$$= \frac{1}{2} v_0 \sum_{k_1 k_3} \hat{a}^+_{k_1}\left(-\hat{a}^+_{k_3}\hat{a}_{k_1}\hat{a}_{k_3}\right) =$$

$$= \frac{1}{2} v_0 \sum_{k_1 k_3} (-)\,\hat{a}^+_{k_1}\left(\delta_{k_1 k_3} - \hat{a}_{k_1}\hat{a}^+_{k_3}\right)\hat{a}_{k_3} =$$

$$= \frac{1}{2} v_0 \sum_{k_3, k_1}\left(\hat{a}^+_{k_1}\hat{a}_{k_1}\hat{a}^+_{k_3}\hat{a}_{k_3} - \hat{a}^+_{k_1}\hat{a}_{k_1}\delta_{k_3,k_1}\right)$$

$$= \frac{1}{2} v_0 \left\{\left(\sum_{k_1}\hat{a}^+_{k_1}\hat{a}_{k_1}\right)\left(\sum_{k_3}\hat{a}^+_{k_3}\hat{a}_{k_3}\right) - \left(\sum_{k_1}\hat{a}^+_{k_1}\hat{a}_{k_1}\right)\right\}$$

$$= \frac{1}{2} v_0 \left\{\left(\hat{N}\right)^2 - \hat{N}\right\}$$ which is correct result.

$\Rightarrow$ Always use normal-ordered form

$$\hat{V} = \frac{1}{2V}\sum_{\substack{k_1 k_2 k_3 k_4 \\ (k_1+k_2=k_3+k_4)}} v(\bar{k}_4-\bar{k}_2)\,\hat{a}^+_{k_1}\hat{a}^+_{k_2}\hat{a}_{k_4}\hat{a}_{k_3}$$